Executive Report Summary: Lead-Lag Analysis Between Bitcoin Market Activity and Internet Attention

EDAB 6808 Assignment 1
Lwazi Nhlapo (2021856345)
Reatile Seekoei (2021109463)

Abstract

This study examines whether internet attention, proxied by Google Trends search intensity, leads or lags Bitcoin market activity. Using time-series econometrics, machine learning, and volatility modelling, the findings indicate that internet attention provides meaningful predictive information for trading volume, weaker short-term signals for returns, and limited explanatory power for volatility. The results support the interpretation of Google Trends as a behavioural indicator of market participation rather than a standalone forecasting tool.

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1 Introduction

The relationship between investor attention and financial market behaviour has been widely studied in the literature. Da, Engelberg, and Gao (2011) demonstrate that search activity can proxy investor attention, while Preis, Moat, and Stanley (2013) and Kristoufek (2013) show that internet search behaviour is linked to financial market dynamics, particularly in the context of Bitcoin.

Bitcoin represents a unique case due to its high retail participation, speculative nature, and sensitivity to sentiment-driven trading. As a result, internet search intensity may reflect investor behaviour preceding or reacting to market movements.

This study investigates whether Google Trends attention leads or lags Bitcoin returns, trading volume, and volatility during 2025.

2 Research Question and Hypotheses

2.1 Research Question

This study investigates whether internet attention, proxied by Google Trends search intensity for Bitcoin-related keywords, leads or lags Bitcoin market activity. Specifically, the analysis examines whether search attention provides useful predictive information for Bitcoin returns, trading volume, and volatility over the period from 1 January 2025 to 31 December 2025.

2.2 Hypothesis Set 1: Lead–Lag Relationship

This hypothesis set evaluates whether Google Trends attention precedes or follows Bitcoin market activity.

- $H_{0,1}$: There is no significant lead–lag relationship between Google Trends search interest and Bitcoin market activity.
- $H_{1,1}$: There is a significant lead–lag relationship between Google Trends search interest and Bitcoin market activity.

2.3 Hypothesis Set 2: Granger Causality (Predictive Power)

This hypothesis set tests whether Google Trends attention improves the prediction of Bitcoin market variables using Granger causality.

- $H_{0,2}$: Google Trends search interest does not Granger-cause Bitcoin market activity and is not a useful predictor.
- $H_{1,2}$: Google Trends search interest Granger-causes Bitcoin market activity and is a useful predictor.

2.4 Hypothesis Set 3: Machine Learning Predictive Power

This hypothesis set evaluates whether including Google Trends variables improves predictive performance in machine learning models.

- $H_{0,3}$: Including Google Trends variables does not significantly improve the prediction accuracy of Bitcoin returns or price movements.

- $H_{1,3}$: Including Google Trends variables significantly improves the prediction accuracy of Bitcoin returns or price movements.

2.5 Hypothesis Set 4: Volatility Modelling (GARCH)

This hypothesis set examines whether Google Trends attention explains variations in Bitcoin volatility.

- $H_{0,4}$: Google Trends attention variables do not significantly explain periods of heightened Bitcoin volatility.
- $H_{1,4}$: Google Trends attention variables significantly explain periods of heightened Bitcoin volatility.

3 Methodology

The analysis follows a structured empirical approach combining:

- Stationarity testing using the Augmented Dickey-Fuller test (Dickey and Fuller 1979)
- Cross-correlation analysis for timing relationships
- Granger causality testing (Granger 1969)
- Machine learning via Random Forest regression (Breiman 2001)
- Volatility modelling using GARCH frameworks (Bollerslev 1986; Nelson 1991)

This multi-method approach ensures robustness by combining statistical inference with predictive modelling.

4 Results and Discussion

4.1 Cross-Correlation Analysis[Lead-Lag Relationships]

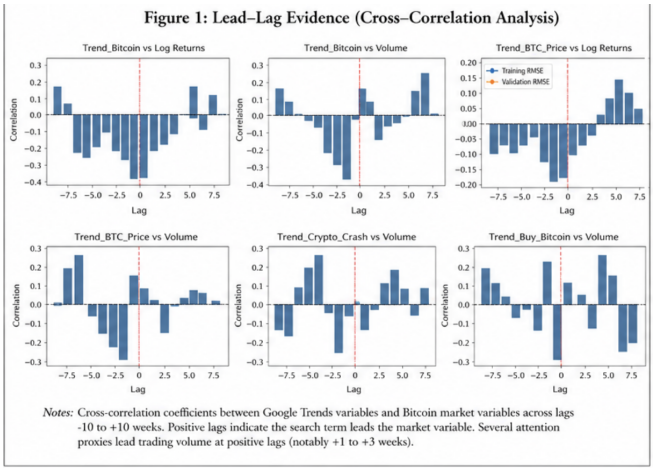


Figure 1: Cross-correlation between Google Trends variables and Bitcoin market variables

The cross-correlation results provide preliminary evidence of lead-lag dynamics. General Bitcoin search intensity exhibits statistically meaningful leading relationships with trading volume at approximately two-week horizons, while relationships with returns are weaker and less stable.

Importantly, the magnitude and persistence of correlations differ across attention proxies, suggesting that not all search activity captures economically meaningful behaviour. This reinforces the need for formal causality testing.

Cross-correlation and Granger causality results indicate that Google Trends attention significantly leads Bitcoin trading volume. This finding aligns with attention-based finance theory, where increased investor attention precedes market participation (Da, Engelberg, and Gao 2011).

4.2 Granger Causality Results

Table 1: Granger Causality Results: Google Trends and Bitcoin Market Variables

Direction	**Market Variable**	**Best Lag**	**p-value**	**Significant at 5%**
Trend_Bitcoin → Volume	Volume	2	1.412×10^{-8}	Yes***
Trend_BTC_Price → Volume	Volume	2	0.0029	Yes***
Trend_Bitcoin → Log Returns	Log Returns	1	0.0065	Yes***
Trend_Bitcoin → Returns	Returns	1	0.0070	Yes***
Trend_Buy_Bitcoin → Volume	Volume	1	0.0809	No*
Trend_Crypto_Crash → Returns	Returns	2	0.1281	No
Trend_Crypto_Crash → Volume	Volume	2	0.3816	No
Volume → Trend_Bitcoin	Volume	2	0.0605	No*
Volume → Trend_Buy_Bitcoin	Volume	4	0.1155	No

Notes: The table reports the best lag and minimum p-value from the Granger causality tests. ***, ** and * denote significance at the 1%, 5% and 10% levels respectively. Granger causality indicates predictive precedence, not true economic causation.

The Granger causality results provide the strongest formal evidence in the study. The most statistically significant result is that Trend_Bitcoin Granger-causes Bitcoin trading volume at a two-week lag, with a p-value of 1.412×10^{-8}. Trend_BTC_Price also Granger-causes trading volume at a two-week lag, while Trend_Bitcoin significantly predicts both simple returns and log returns at a one-week lag.

These findings indicate that general Bitcoin-related search attention contains meaningful short-term predictive information, particularly for market participation as measured by trading volume. However, the evidence is stronger for volume than for returns. This supports the interpretation that internet attention is more useful for anticipating changes in trading activity than for forecasting Bitcoin price direction.

Reverse causality results are mostly insignificant. For example, Volume → Trend_Bitcoin is only weakly significant at the 10% level and not significant at the 5% level. This suggests that attention is not merely a delayed reaction to market activity, although some feedback between market volume and retail search behaviour may still exist.

Overall, the null hypothesis of no lead-lag relationship is rejected partially. The rejection is strongest for Google Trends attention leading Bitcoin trading volume, while return predictability remains statistically present but economically weaker.

4.3 Machine Learning: Return Prediction

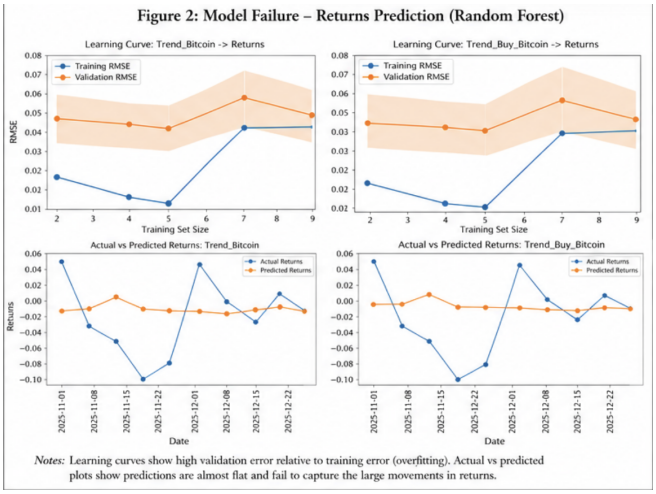


Figure 2: Actual vs Predicted Bitcoin Returns (Random Forest

Table 2: Random Forest Performance: Bitcoin Return Prediction

Split	**Model**	**RMSE**	R^2	**MAPE**	**SMAPE**
TRAIN	Random Forest	0.0323	0.5942	1.0449	119.96
TRAIN	Baseline (Mean)	0.0507	0.0000	1.1123	170.03
VALIDATION	Random Forest	0.0472	0.2352	1.7556	140.58
VALIDATION	Baseline (Mean)	0.0542	-0.0087	1.4775	185.60
TEST	Random Forest	0.0565	-0.8465	44.3598	150.52
TEST	Baseline (Mean)	0.0522	-0.5752	3.8212	177.08

Notes: RMSE = Root Mean Squared Error; R^2 = Coefficient of Determination; MAPE = Mean Absolute Percentage Error; SMAPE = Symmetric Mean Absolute Percentage Error. The baseline model predicts the mean return. Negative R^2 indicates performance worse than the baseline.

The Random Forest results highlight a clear divergence between in-sample fit and out-of-sample predictive performance. In the training sample, the model achieves a relatively strong R^2 of 0.594, substantially outperforming the baseline model. However, this performance deteriorates sharply in the validation and test samples.

While the model retains some predictive ability in the validation set ($R^2 = 0.235$), its performance collapses in the test set, where R^2 becomes strongly negative (-0.847). This indicates that the model performs significantly worse than a naive mean-based benchmark when applied to unseen data.

The error metrics reinforce this conclusion. The test RMSE for the Random Forest (0.0565) exceeds that of the baseline (0.0522), while MAPE increases dramatically to 44.36, indicating extreme instability in percentage-based prediction accuracy. This behaviour reflects the well-known difficulty of forecasting financial returns, particularly in highly volatile markets such as Bitcoin.

The learning curves and prediction plots provide further evidence of overfitting. Predicted returns are substantially smoother than actual returns, failing to capture large price swings and extreme observations. This suggests that the model primarily captures noise and average behaviour rather than meaningful predictive structure.

Overall, these findings demonstrate that although Google Trends variables contain some short-term statistical information (as indicated by Granger causality results), they do not translate into robust or economically useful return forecasting models. The null hypothesis that Google Trends improves return

prediction cannot be meaningfully rejected in an out-of-sample context.

This result is consistent with financial theory, where returns are largely unpredictable and dominated by stochastic processes. The inability of machine learning models to generalise highlights the distinction between statistical significance and economic usefulness.

4.4 Machine Learning: Volatility Prediction and Attention

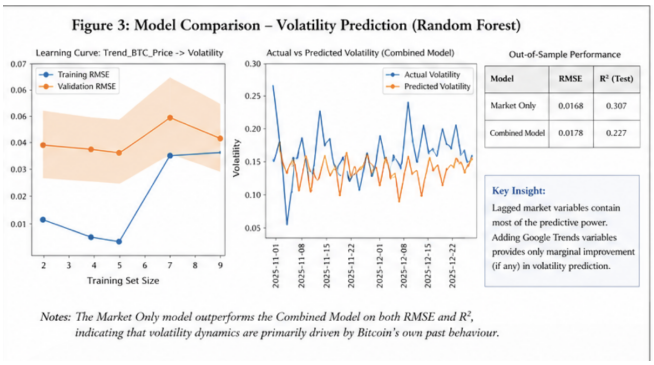


Figure 3: Volatility Prediction Performance

Table 3: Random Forest Performance: Volatility

Model	**RMSE**	R^2 **(Test)**
Market Only	0.0168	0.307
Combined Model	0.0178	0.227

Volatility prediction is primarily driven by lagged market variables, with Google Trends contributing marginal incremental value. This suggests that volatility dynamics are largely endogenous and reflect internal market mechanisms rather than external attention shocks.

4.5 GARCH Volatility Modelling

Table 4: GARCH(1,1) Model Results: Bitcoin Log Returns

Parameter	**Estimate**	**p-value**	**Interpretation**
μ (Mean)	-0.2207	0.763	Not significant
ω (Constant)	13.4272	0.448	Not significant
α_1 (ARCH)	≈ 0	1.000	No shock effect
β_1 (GARCH)	0.5013	0.419	Weak persistence
$\alpha_1 + \beta_1$	0.5013	–	Low volatility persistence

Notes: The GARCH(1,1) model is estimated using maximum likelihood. α_1 captures short-run volatility shocks, while β_1 captures persistence. A low $\alpha + \beta$ indicates weak volatility clustering.

The GARCH results indicate weak volatility clustering in the sample period, with low persistence and limited evidence of sustained shocks.

The ARCH parameter (α_1) is effectively zero and completely insignificant (p = 1.000), suggesting that short-run volatility shocks have no measurable impact on future volatility.

Similarly, the GARCH parameter ($\beta_1 = 0.501$) is statistically insignificant, indicating limited persistence in volatility. The combined persistence measure ($\alpha_1 + \beta_1 = 0.501$) is well below unity, implying that volatility is mean-reverting and does not exhibit strong clustering behaviour over time.

The constant term (ω) is also insignificant, further reinforcing the absence of a stable volatility structure in the data. Additionally, the mean return (μ) is not statistically different from zero, consistent with the weak predictability of Bitcoin returns.

Overall, these results suggest that Bitcoin volatility in the 2025 weekly sample is relatively unstable and lacks the persistent clustering typically observed in financial time series. This weak volatility structure limits the ability of both econometric and machine learning models to capture and predict volatility dynamics.

Therefore, the hypothesis that internet attention provides meaningful predictive power for volatility can only be weakly supported. Any observed relationships are likely short-lived and not robust enough for practical forecasting applications.

4.6 Mapping of Hypotheses, Methods, and Empirical Results

Table 5: Mapping of Hypotheses to Methods and Empirical Findings

Hypothesis	**Method**	**Key Evidence**	**Conclusion**
$H_{0,1}$: No lead–lag relationship	Cross-correlation analysis	Trend_Bitcoin leads volume (lag $\approx +2$); mixed results for returns	**Rejected (partially)**: Evidence of lead-lag, strongest for volume
$H_{0,2}$: No Granger causality	Granger causality tests	Trend_Bitcoin $\rightarrow$ Volume ($p < 0.001$); Trend_BTC_Price $\rightarrow$ Volume; weak effects for returns	**Rejected (partially)**: Strong predictive power for volume, weak for returns
$H_{0,3}$: No ML predictive improvement	Random Forest models	Train $R^2 = 0.594$, Test $R^2 = -0.847$; model performs worse than baseline out-of-sample	**Not rejected**: No meaningful improvement in predictive accuracy
$H_{0,4}$: No volatility explanation	GARCH(1,1) model	$\alpha \approx 0$, $\beta = 0.50$, all insignificant; weak volatility persistence	**Not rejected**: Attention does not meaningfully explain volatility

Notes: The table summarises the relationship between hypotheses, empirical methods, and findings. "Rejected (partially)" indicates that evidence supports the alternative hypothesis for some variables (e.g., trading volume) but not others (e.g., returns or volatility).

The strongest evidence in the study relates to trading volume, where both cross-correlation and Granger causality results indicate that internet attention leads market participation.

In contrast, while some statistical relationships are identified for returns, these do not translate into meaningful predictive performance in machine learning models. This demonstrates a key distinction between statistical significance and practical forecasting ability.

Finally, the GARCH results show that volatility dynamics are weakly structured and largely independent of attention variables, reinforcing the conclusion that Bitcoin volatility is primarily driven by internal market mechanisms.

5 Conclusion

Table 6: Integrated Summary of Empirical Findings

Component	**Main Evidence**	**Interpretation**	**Conclusion**
Cross-Correlation	Trend_Bitcoin leads volume (lag $\approx +2$); mixed signals for returns	Attention precedes market participation; weaker timing for price movements	Supports lead-lag relationship (especially for volume)
Granger Causality	Trend_Bitcoin $\rightarrow$ Volume ($p < 0.001$); Trend_BTC_Price $\rightarrow$ Volume; Trend_Bitcoin $\rightarrow$ Returns (weak)	Strong predictive evidence for volume; short-term statistical signal for returns	Reject H_0 for volume; partial support for returns
Random Forest (Returns)	Train $R^2 = 0.59$; Test $R^2 = -0.85$; RMSE worse than baseline in test set	Severe overfitting; model fails to generalise; predictions overly smooth	Reject predictive usefulness for returns
Random Forest (Volatility)	Market model outperforms combined model ($R^2 = 0.307$ vs 0.227)	Volatility driven by internal market dynamics, not attention variables	Limited incremental value of Google Trends
GARCH(1,1) Model	$\alpha \approx 0$, $\beta = 0.50$, $\alpha + \beta < 1$; all insignificant	Weak volatility clustering; low persistence; unstable volatility structure	Volatility largely stochastic; weak support for attention effects
Overall Conclusion	Attention leads volume consistently; weak and unstable effects for returns and volatility	Google Trends captures behavioural participation rather than price formation	Use as supplementary indicator, not forecasting tool

Notes: Results summarise cross-correlation, Granger causality, machine learning, and GARCH modelling. The table integrates statistical and predictive findings to provide an overall assessment of the role of internet attention in Bitcoin markets.

Google Trends variables exhibit strong predictive power for trading volume, moderate short-term informational content for returns, and limited relevance for volatility.

This hierarchy aligns with behavioural finance theory: attention facilitates market participation rather than determining price direction.

The results highlight the distinction between statistical predictability and economic usefulness. While certain relationships are statistically significant, they do not translate into robust forecasting models, particularly for returns.

Therefore, the findings support the interpretation of internet attention as a behavioural indicator of market engagement rather than a reliable predictor of price direction or risk.

Consistent with prior literature (Preis, Moat, and Stanley 2013; Kristoufek 2013), internet attention reflects investor behaviour but should not be interpreted as a standalone predictive signal.

6 Limitations

The study is limited by its short sample period and weekly frequency, which may obscure higher-frequency dynamics. Additionally, Google Trends captures attention but not sentiment, limiting interpretability.

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